



Power Academy

RDBMS Sales Training

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Computer Associates

- Second largest software company behind Microsoft -- \$2.74B
- 22.2% net profit margin -- best in database industry
- Primary segments are applications, system software, systems management
- Client-server is now 30% of business; remainder is mainframe
- Legent acquisition adds more systems management focus and pizzazz to CA-Unicenter
- Database products include IDMS, Datacom, and Ingres
- More open, more innovative but still selling around negative customer perceptions

CA-Ingres 1996 Themes

- **“We haven't died! But we were acquired by Computer Associates. And we are behind Oracle, Sybase, and Informix.”**
 - **New versions of OpenIngres RDBMS and OpenRoad tool**
 - **Excellent connectivity into IBM MVS applications**
 - **Higher product quality, service offerings**
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- **Ingres should not be ignored, but CA with Ingres is not much of a market factor today outside the installed base**

IBM DB2

- Typically a bundled sell with other IBM products, especially hardware and services
 - Extensive, worldwide service includes outsourcing and system integration
- The future for DB2:
 - Add more non-IBM platforms
 - Match/surpass the independents in RDBMS technology
 - Scale from high-end SMP, clusters to MPP SP2
 - Better coordination between DB2 versions

IBM 1996 Themes

- Build towards “best in class” image with best price/performance
- Make the SP the market leader for mission-critical applications, particularly data warehousing
- DB2/6000 version 2 brings new, leading levels of functionality
- Continue roll-out of RS/6000 SMP servers “the second generation”
- Innovative query tools for data analysis and complex decision support = Data Mining

Informix

- **Service and support receive good marks from users but is limited in scope**
 - Informix service revenues 25%
 - Oracle service revenues 52%
- **The future:**
 - Improve parallel-scalable technology with clusters and MPP soon-shipping Version 8
 - Competitive replication technology
 - Deliver OO DB framework in 1996
 - Achieve better sales for top-notch NewEra OO-GUI development tool

Informix 1996 Themes

- **OnLine 7.1 for mass-market SMP servers**
- **OnLine 8.0 rollout for MPP, clusters**
- **NewEra for enterprise-class, client-server development**
- **Store-anything framework ... post-relational**
- **Lower priority**
 - **Workgroup/OnLine for NT, Personal/SE**
 - **Replication**
 - **Systems Management**
 - **Gateways via DRDA**

Oracle

- **New focus on services**
 - Services now exceed product revenues
 - Vertical-industry services (telecomm, oil/gas, etc.)
 - Sales reps carry a consulting quota
- **The future:**
 - Competition with IBM, Microsoft and CA for overall IT market share
 - Internet, multimedia set-top boxes and Hollywood
 - Object-oriented technology in Oracle 8

Oracle 1996 Themes

- Oracle 7.3 is still the performance champion
- Enterprise Scalability, one-stop-shopping
- You need consulting ... we have it
- Applications business: turn a liability into an asset
- Workgroup 2000
 - Personal, WG, wireless, easy to install/administer
- Multimedia platform-of-choice (e.g., US West)
- Oracle 8 -- Sorry, not until 1997

Sybase

- **Second-largest independent RDBMS supplier**
 - Past 60-80% growth rate
 - Powersoft acquisition beefs up tools
 - Mostly direct sales, few ISVs and apps
- **The future (System 11 is *not* just another release):**
 - Add parallel-scalable technology, shared-less
 - “Leading-edge” distributed database management on Tivoli framework
 - Merge Navigation Server data warehousing, sell IQ Accelerator for warehousing

Sybase 1996 Themes

- **System 11 fixes our performance problems. Period.**
- **Replication -- we have it; you need it**
- **Powersoft 5.0 improves a leading toolset**
- **IQ Accelerator -- yes, we do complex decision support**
- **Navigation Server -- yes, we do parallel MPP**
- **Regain Wall Street superstar status**

Microsoft

- **Windows NT Server moves Microsoft into the department and soon the distributed enterprise**
- **SQL Server 6.0**
 - **Highly aggressive pricing**
 - **All-Microsoft code developed by a world-class team**
 - **Tight integration between OS and RDBMS**
 - **Excellent performance**
 - **Intel/Microsoft represent a definite threat to uniprocessor RS/6000**

Microsoft

- **Service and support less extensive than other suppliers. Few people & programs.**
- **The future:**
 - **Scale towards the glass house in 1997**
 - **Catch up with RDBMS technology leaders**
 - **Grow market share**
 - **Add support channels, applications**
- **Challenge: gaining the trust of IT executives for enterprise-class computing assignments**

Progress Software

- Progress plays to the tune of a different drummer
 - Avoids head-to-head battles with RDBMS leaders
 - Sells effectively through VAR/ISV channel, has little direct sales
 - Technology works for the workgroup/departmental applications they target
 - RDBMS scales modestly, lacks pizzaz features
 - Toolset is quite competent for Windows apps
- Selling Tips
 - Qualify on need/desire for name-brand RDBMS, app fit
 - VAR/ISV can benefit from IBM's customer experience
 - Best fit is RS/6000 uniprocessor or small SMP

Red Brick

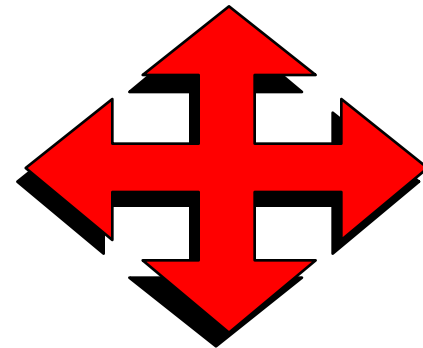
- **Specialist in RDBMSs for decision support**
- **Uses pre-calculated Joins to make queries fast, but Inserting is slow due to complex index building**
- **Small (struggling) company, but focused on Data Warehouses, the hot data management topic**
- **Selling Tips**
 - **Qualify on customer willingness to buy from small co.**
 - **Don't even think about Red Brick for an OLTP app**
 - **Red Brick and IBM are best together for Data Warehouse on RS/6000 SMP**

Software AG's Adabas D

- SAG is long-setablished and respected on S/390
- Adabas D on RS/6000 is top-selling platform for SAG
- “Windows NT Pricing Model” = low-cost proposals
- Tied with Informix for #2 RDBMS on SAP applications
- Interesting Features
 - Location transparency; a federated database
 - Distributed referential integrity
 - NIST 127-2 secure database certification
- Good performance and scalability at low cost

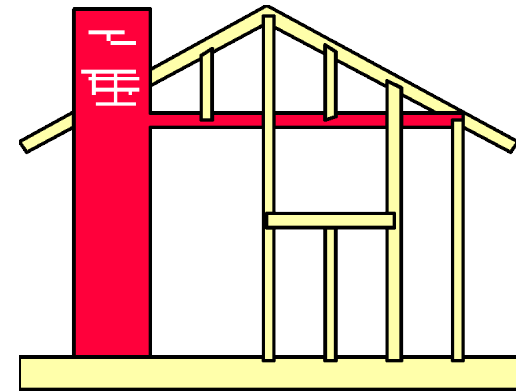
Aberdeen's 1996 RDBMS Themes

- “Mine is better than yours”
- Store anything
- WG, Personal and wireless
- Replication = distributed
- More OLTP performance
- TPC-D = decision support
- Better administration
- Tool wars, round 3



Aberdeen's 1996 RDBMS Rating Categories

- Scalability
- Distributed Data
- Open Data Access
- Development Tools
- Other Technology
- End-User Solutions

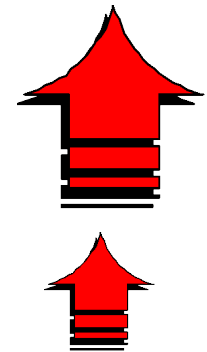


*Source: Aberdeen's Distributed, Open Production Systems
Buying Guide*

Scalability

Can I write an application once and have it grow or shrink to fit changing enterprise needs?

- Non-parallel performance
- Parallel performance
- Large databases
- High-availability features



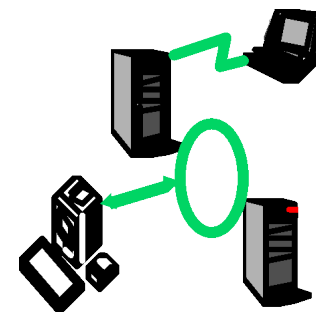
Scalability Notes

- CA Ingres -- Modest scalability; no MPP
- IBM DB2 for AIX
 - Competitive but not leading on SMP
 - *Only* compatible SMP and MPP story except AT&T
- Informix
 - Good scalability to 8 processors on SMP. Many TPC-Cs.
 - Now on SP2 and other loosely-coupled HW with version 8
- Microsoft -- Modest scalability; no MPP; good on Alpha
- Oracle
 - Good scalability story from desktop to SP2 & ES/9000
 - No TPC-C benchmarks until July 1995 -- Hot stuff
- Sybase
 - System 11 improves scalability markedly. Good on uni.
 - Navigation Server: Poor on AT&T 3600 should get better in 1996 on RS/6000 SP

Distributed Technology

Can I move computing and data out to workers and still be able to effectively administer it?

- Tightly-coupled data -- two-phase commit
- Loosely-coupled data -- replication
- Simultaneous-update data -- peer-peer
- Distributed administration



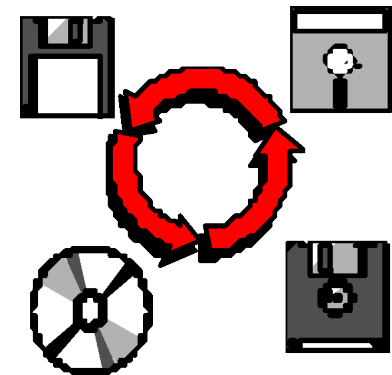
Distributed Technology Notes

- **CA Ingres:** Was best functionality; now adequate
- **IBM DB2 for AIX:** Good async replication in Data Propagator, but no peer-peer replication
- **Informix:** Adequate, including peer-peer
- **Microsoft:** Limited replication in SQL Server 6 for NT
- **Oracle:** Good replication including peer-peer
- **Sybase:** OK asynchronous replication; no peer-peer; good heterogeneous replication

Open Technology

How will my applications work with other data sources?

- Standards
- Gateways, open clients
- Open OLTP monitors
- Access to other supplier's RDBMSs



Open Technology Notes

- **CA Ingres:** good mainframe gateways to ES/9000 DBMS
- **IBM DB2 for AIX:** full 2-way DRDA, ODBC; limited non-IBM gateways. Best choice in IBM homogeneous world.
- **Informix:** DRDA client; limited gateways, but IBI EDA/SQL ties. Improving open APIs
- **Microsoft:** great for PC apps; OLE; ODBC inventor
- **Oracle:** DRDA client; broad gateways. Improving APIs.
- **Sybase:** Open Client & Open Server appeals to 3GL bigots. Best gateway breadth. Overall, most flexible.

Distributed Tool Technology

How can I efficiently develop, deploy and maintain distributed applications?

- Distributed application support
- Open technology
- Programmer technology
- Other development technology

Development Tool Notes

- **CA Ingres:** OpenROAD is most mature OO CADE
- **IBM DB2 for AIX:** No comparable tools for OO CADE. More limited ISV selection than Informix, Oracle and Sybase
- **Informix:** New Era is maturing as OO CADE
- **Microsoft:** MFC as a 4GL is excellent for MS environment
- **Oracle:** clunky tools only work on Oracle. Oracle Objects is interesting but low-end, incompatible with Developer/2000. Sedona: True OO-Forms replacement in 1997.
- **Sybase:** Powersoft & Watcom are among market leaders, but PB needs better scalability for enterprise apps

Other Technologies

What other technologies are needed to develop superior distributed applications?

- **Business-rules processing**
- **BLOBS, multimedia**
- **Non-standard data types**
- **Object extensions**

Other Technology Notes

- **CA Ingres:** partnership with Fujitsu for 1st ORDBMS in '96
- **IBM DB2 for AIX:** fast moving DB2 into searchable audio, video, user-defined data types. Hot stuff.
- **Informix:** Practical BLOB support today. In 1996, OO framework, same searchable data types as IBM
- **Microsoft:** OLE, futures
- **Oracle:** Most aggressive in multimedia marketing; objects in 1997; serious about video on demand
- **Sybase:** more multimedia/OO talk than action

Supplier Solutions

What else can my RDBMS supplier bring to the table?

- Applications
- Third-party support
- Consulting & services

Supplier Added-Value Notes

- **CA Ingres:** almost no services; many ISVs plus CA apps
- **IBM DB2 for AIX:** world-class, worldwide services; growing number of ISVs with many apps
- **Informix:** low services; many ISV apps; never own apps
- **Microsoft:** no services; most apps are for workgroups
- **Oracle:** tons of services; many ISVs; Oracle applications
- **Sybase:** fewer services; fewer ISVs; no applications

RDBMS Supplier Weaknesses

Nobody's Perfect

CA-Ingres Weaknesses

- RDBMS architecture is behind -- no parallel/scalable
- No Ingres people remain -- but few customers have defected
- CA sales force does not understand the Unix world well
- Product quality was poor; now much better
- OPENROAD left customers with character 4GL behind -- totally different, incompatible environment
- Hard to sell when you're number four and trailing

IBM DB2/6000 Weaknesses

- Perception is DB2/6000 is not a world-class product
 - Old limits are still remembered.
 - Must shatter perceptions of limited scalability
- Limited cross-platform, cross-OS capabilities & support
- Still caution by "open" zealots
- Note: These weaknesses are relative. DB2/6000 is now a competitive product.

Informix Weaknesses

- No-consulting policy slowed large enterprise growth
- Late with desktop & workgroup database/tools
- NewEra 2.0 makes the product more useable
- Behind in replication, fair in gateways
- Slow to build direct sales force, marketing ... missed growth opportunities over past two years
- No NetWare story — less a problem going forward

Oracle Weaknesses

- RDBMS engine looks dated -- is version 7.3 a TPC-C stop-gap or a major re-write?
- No 2nd generation client-server tools, so-so on Windows
- Service/consulting drive makes ISV partnering tough
- Deal-driven, not partnership driven
- Unfocused: Hollywood multimedia, applications, TV set-tops
- Note: Oracle has never been stronger in the 1990s than it is now

Sybase Weaknesses

- 1980s architecture runs out of steam -- they need System 11 *now*
 - Poor scalability in SMP
 - database size = departmental RDBMS
- Poor GUI tools drove Powersoft acquisition
- Navigation Server is platform-limited to the NCR 3600, an OLTP dog (but SP2 version is imminent)
- Few application-solution partners & solutions
- Very high sales force turnover in 1995
- Over-reaching sales tactics turn off customers
- Momentum is downwards but could recover in 1996

Where is IBM RS/6000 Best Positioned Today?

- **Worldwide support and service ... the SAP R/3 story**
- **Enterprises that commit to DB2 across IBM platforms**
- **With mainframe ISVs who have ported to client-Unix-server**
- **Departmental systems to 4 processors for OLTP**
- **Expandable workgroup-plus servers above/at OS/2 level**
 - **SMP & SP improve ability to do bigger systems**
- **Modest-size data warehouses on SMP; larger on SP**
 - **Products now in place. Field experience will come over next year. MCI is a good story.**

IBM & RDBMS Partners: Partnership Propositions

Partner	Partnership Proposition
Computer Associates' Ingres	Distributed systems for downsizing enterprises
Informix	RS/6000 & Informix: great performance and price/ performance
IBM RS/6000 and DB2/6000	Optimized for each other; one-stop shopping; one RDBMS
Oracle	Market leaders in unifying the enterprise from desktop to datacenter
Sybase	Best performance on RS/6000; PowerBuilder; IQ & Nav Server

ISV Support for Unix Platforms

- Application solutions pull through hardware/RDBMS
- IBM gained market share in 1994, usually at the expense of HP

Platform Choices by Hardware Supplier

Source: Aberdeen Group and ISVs

ISV	HP	IBM	Sun	DEC	AT&T	Other
BGS Systems	30%	60%	10%			
Cincom Systems	35%	35%	20%			10%
Genesys Software	30%	50%	20%			
Information Builders	25%	25%	25%		20%	5%
Landmark Systems	20%	40%	40%			
CA Unicenter	40%	35%	15%	2%	4%	4%
Cyborg	30%	20%	10%	10%		30%
Dun & Bradstreet	50%	30%	15%			50%
IRI Software	38%	25%	14%	15%	4%	4%
Lawson Software	50%	30%	10%			10%
SAP America	35%	22%	5%	11%		27%
SAS Institute	28%	20%	42%	4%		6%
Software AG	50%	30%	15%			5%
SPSS	40%	25%	35%			

Where RS/6000 with DB2 is Best Suited

- **Product Strengths**
 - DB2 for AIX 2 engine -- functionally much improved but nobody knows about it
 - Distributed technology using Data Propagator between mainframe and middle-server RS/6000s
 - Good performance and price-performance
 - Outstanding OLTP support with CICS, Ensina, & Tuxedo
 - Promising scalability with HACMP and SP2
- **Homework: rehearse your CIO and CEO “Elevator Story”**

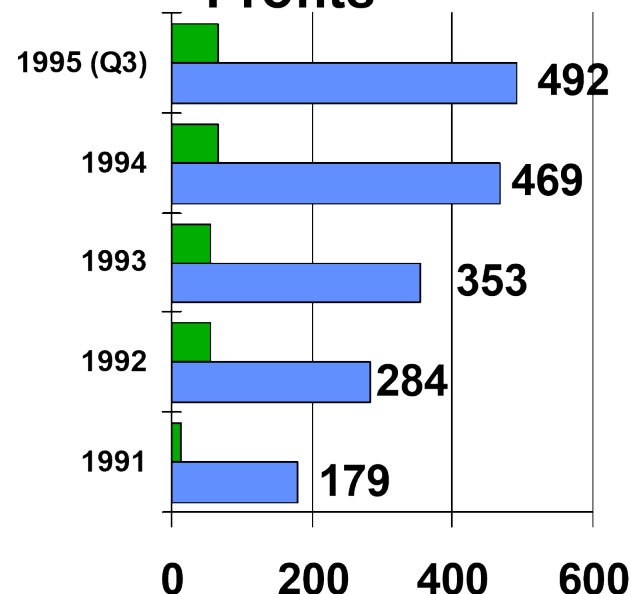


Backup Material

Informix

- Total 1994 corporate revenue equals \$469 million, up 33% from 1993; last 12 months at \$568 million
- Market value \$4.3 billion, up from \$1.5B in 1994
- Share prices up 207% in last 52 weeks!

Informix Revenues & Profits

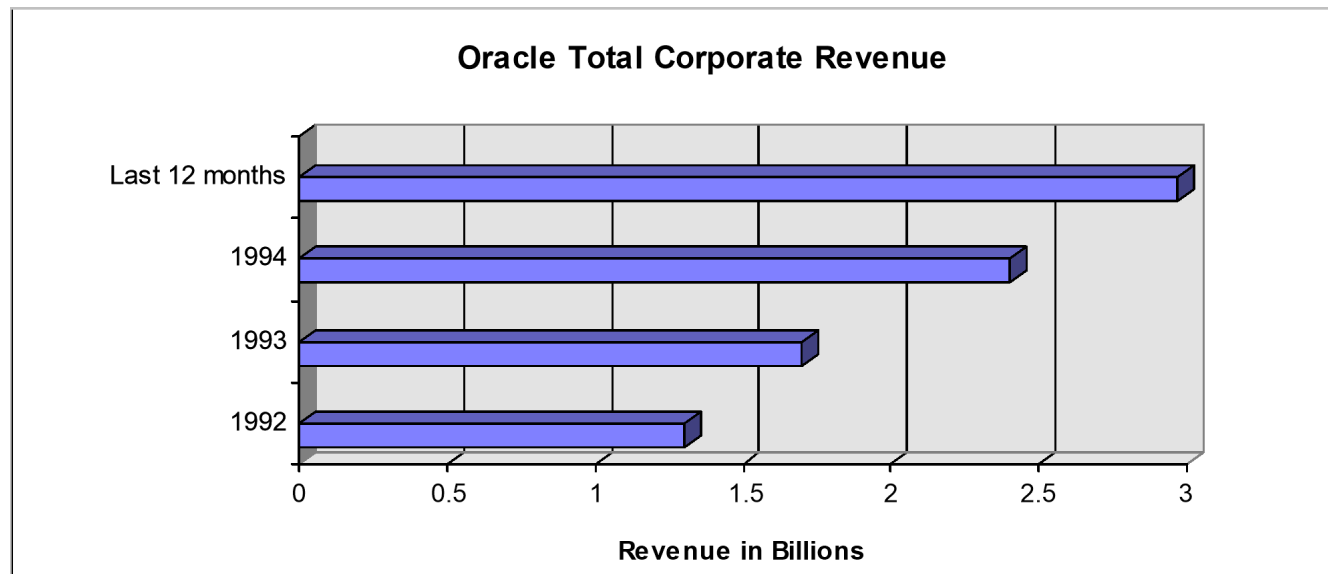


Informix

- **Started with medium-scale & departmental, now scaling to enterprise cluster/MPP with version 8, XPS**
 - **Focus on Unix, indirect channels**
 - **Leader in parallel-scalable (Dynamic Server Architecture)**
 - **NewEra C-S toolset has second-generation technology**
- **Third-largest RDBMS supplier**
 - **Growth rate acceleration — now the fastest growing**
 - **Expanding high-end direct sales with success of DSA in Unix/OLTP and data warehousing as a hot-box**

Oracle

- 1994 total corporate revenue equals 2.4 billion, up 40% from 1993
- Market value \$18.7 billion, up 100% from 1994
- Stock up 134% in last 52 weeks (as of July 29, 1995)



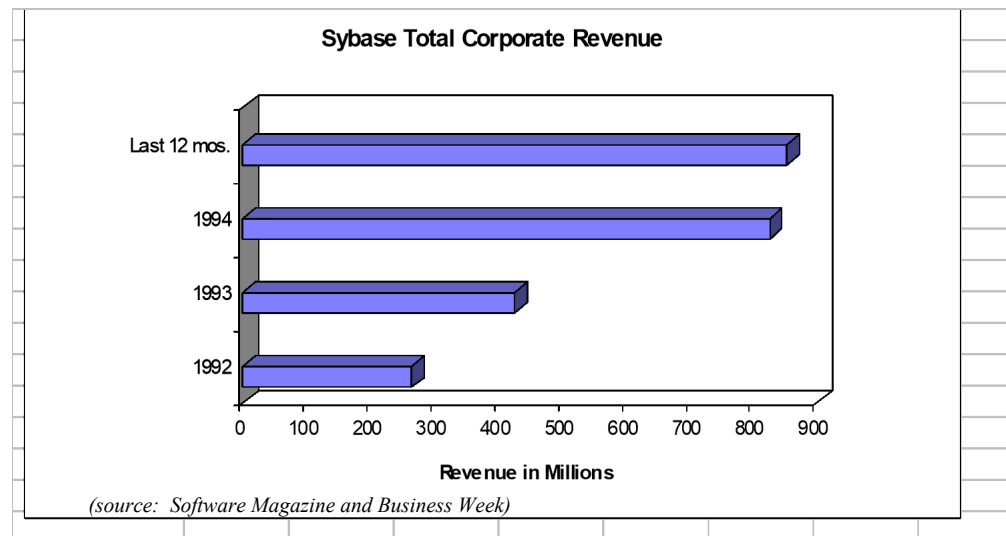
(source: Software Magazine and Business Week)

Oracle

- **Focus on high end & enterprise**
 - Oracle used to sell “runs on any platform”
 - Now sells “broadest functionality, one-stop shopping”
 - New multimedia, text, desktop, and workgroup server products
- **Largest independent RDBMS supplier**
 - On track for \$3.6+ billion in revenues in fiscal 1996
 - Direct sales, but increasing indirect selling

Sybase Inc.

- Total 1994 corporate revenue \$826 million, a 71% increase from 1993
- YTD Sales growth rate 51.9%
- Market Value \$2.4 billion (July 28 1995)
- 41% Stock price drop -- first week April



Sybase

- **Started with medium-scale & departmental, now scaling to enterprise**
 - **“Leading-edge” reputation – fading**
 - **Good replication and connectivity**
 - **Strong fixation on competition with Oracle**
 - **New decision support, workgroup products positioned against Oracle**
 - **Advertising, sales force head-to-head focus**